

Annelids

Annelida means “little rings,” referring to the annelid body’s resemblance to a series of fused rings. Annelids are segmented worms that live in the sea, in most freshwater habitats, and in damp soil. Annelids, which have a coelom (and no hemo-coel), range in length from less than 1 mm to more than 3 m.

Traditionally, the phylum Annelida was divided into three main groups, Polychaeta (the polychaetes), Oligochaeta (the oligochaetes), and Hirudinea (the leeches). The names of the first two of these groups reflected the relative number of chaetae, bristles made of chitin, on their bodies: polychaetes (from the Greek *poly*, many, and *chaitē*, long hair) have many more chaetae per segment than do oligochaetes.

However, a 2011 phylogenomic study and other recent molecular analyses have indicated that the oligochaetes are a subgroup of the polychaetes, making the polychaetes (as defined morphologically) a paraphyletic group. Likewise, the leeches have been shown to be a subgroup of the oligochaetes. As a result, these traditional names are no longer used to describe the evolutionary history of the annelids. Instead, current evidence indicates that the annelids can be divided into two major clades, Errantia and Sedentaria—a grouping that reflects broad differences in lifestyle.

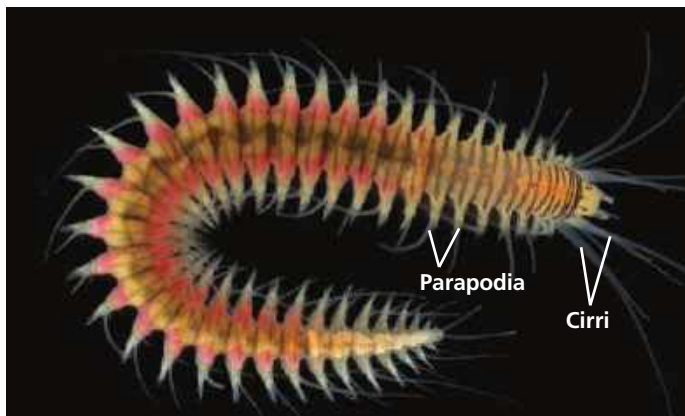
Errantians

Clade Errantia (from the Old French *errant*, traveling) is a large and diverse group, most of whose members are marine (Figure 33.22). As their name suggests, many errantians are mobile; some swim among the plankton (small, drifting organisms), while many others crawl on or burrow in the seafloor. Many are predators, while others are grazers that feed on large, multicellular algae. The group also includes some relatively immobile species, such as the tube-dwelling *Platynereis*, a marine species that recently has become a model organism for studying neurobiology and development.

In many errantians, each body segment has a pair of prominent paddle-like or ridge-like structures called parapodia (“beside feet”) that function in locomotion (see Figure 33.22).

▼ **Figure 33.22** An errantian, the predator *Nereimyra punctata*.

This marine annelid ambushes prey from burrows it has constructed on the seafloor. *N. punctata* hunts by touch, detecting its prey with long sensory organs called cirri that extend from the burrow.



Each parapodium has numerous chaetae. (Possession of parapodia with numerous chaetae is not unique to Errantia, however, as some members of the other major clade of annelids, Sedentaria, also have these features.) In many species, the parapodia are richly supplied with blood vessels and also function as gills. Errantians also tend to have well-developed jaws and sensory organs, as might be expected of predators or grazers that move about in search of food.

Sedentarians

Species in the other major clade of annelids, Sedentaria (from the Latin *sedere*, sit), tend to be less mobile than those in Errantia. Some species burrow slowly through marine sediments or soil, while others live within tubes that protect and support their soft bodies. Tube-dwelling sedentarians often have elaborate gills or tentacles used for filter feeding (Figure 33.23).

Although the Christmas tree worm shown in Figure 33.23 once was classified as a “polychaete,” current evidence indicates it is a sedentarian. The clade Sedentaria also contains former “oligochaetes,” including the two groups we turn to next, the leeches and the earthworms.

Leeches Some leeches are parasites that suck blood by attaching temporarily to other animals, including humans (Figure 33.24), but most are predators that feed on other

▼ **Figure 33.23** The Christmas tree worm, *Spirobranchus giganteus*. This sedentarian’s two tree-shaped whorls are tentacles, which it uses in gas exchange and to collect food particles from the water. The tentacles emerge from a calcium carbonate tube secreted by the worm that protects and supports its soft body.



► **Figure 33.24** A leech. A nurse applied this medicinal leech (*Hirudo medicinalis*) to a patient’s sore thumb to drain blood from a hematoma (an abnormal accumulation of blood around an internal injury).

